HALALAN REVAMP

System Requirements

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Client: University Computer Center

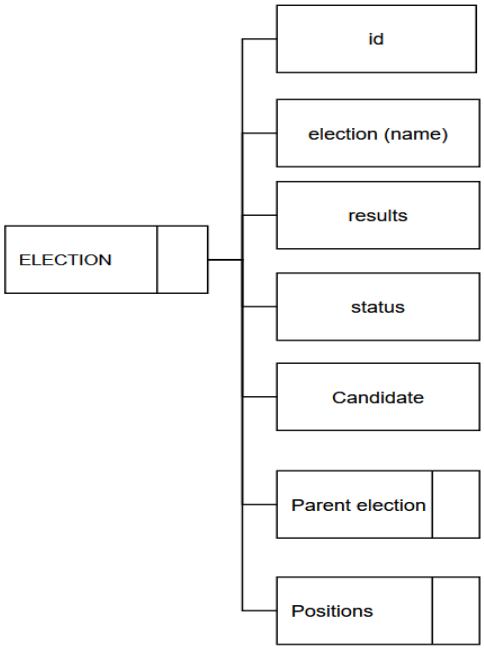
https://cs191-ame.github.io/blog/

CS 191 WFR1
Prof. Ligaya Leah Figueroa

1. Web App Requirements Model

1.1 Content Model

The data tree for the Election component is shown below



1.2 Interaction Model

Actors and Goals

For this election system, 3 main actors have been identified: the administrators (admins), the voters, and external authenticators. The administrators and the voters are primary actors, while the external authenticators are secondary actors.

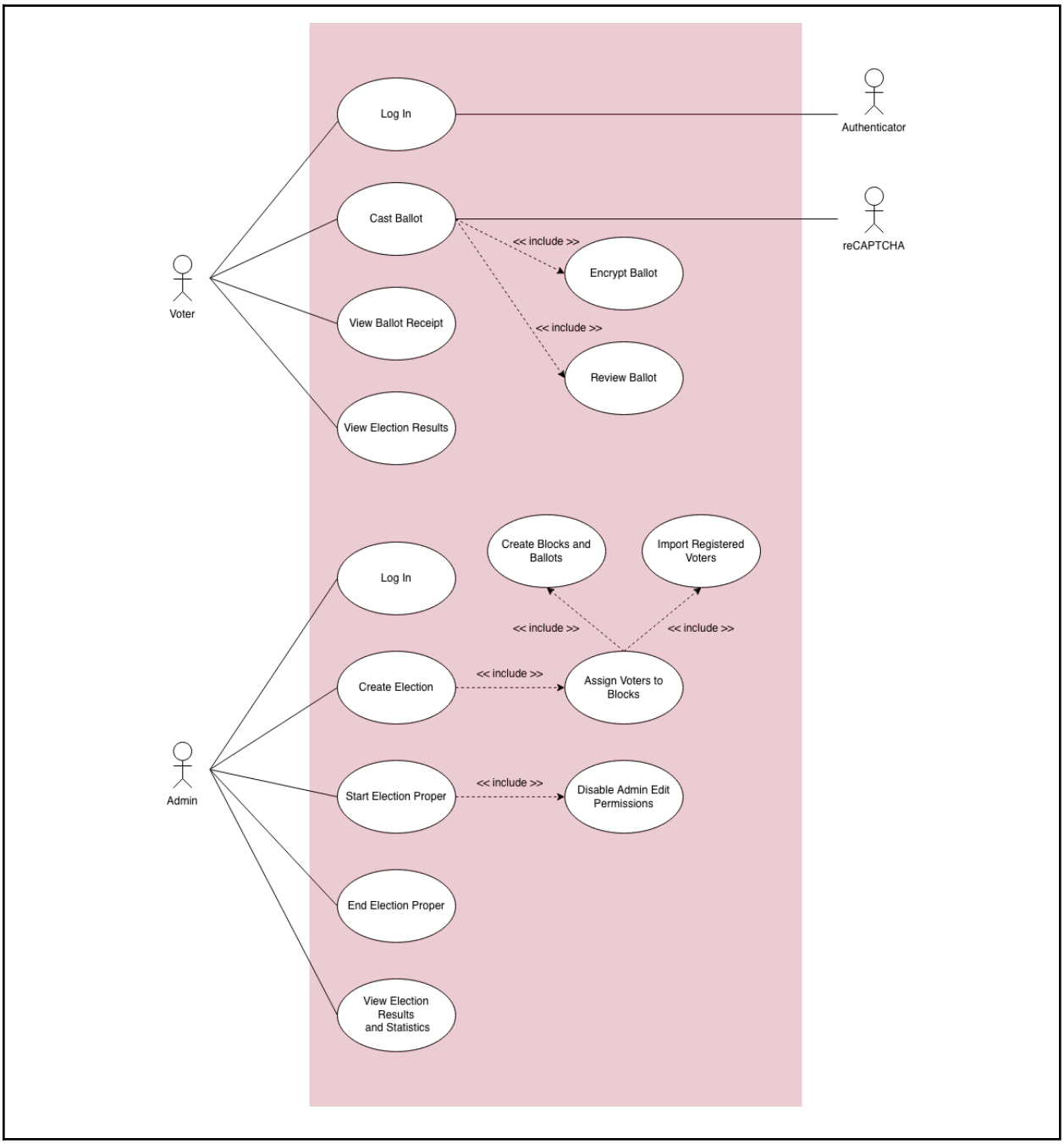
The admins are individuals in charge of managing the elections. The main admins are personnel from the University of the Philippines Computer Center. They will be responsible for the creation of the elections, for compiling and uploading into the system all relevant, and overseeing the election proper. For the university-wide elections, admin status may be extended

to select individuals from each college or department so that they may manage the start and duration of their respective elections. They are considered primary actors as they will be initiating the various events that occur in the system.

The voters consist of members of the university that are eligible and registered to vote in the elections. During the election proper, they are to first, login to the system using their university-issued email address (UP email) and have their identity verified using 2-factor authentication. Then, they may fill out their ballots by selecting candidates. Once they are ready to submit and their vote has been validated by the system, they will be presented with a summary of their vote alongside a prompt to finalize their ballot. Upon submission, they will need to successfully answer a reCAPTCHA for the system to fully process their cast ballot. Voters are a primary actor as well since their actions drive the interactions with the system.

Authenticators such as Google Authenticator and Google reCAPTCHA on the other hand are secondary or participating actors in the system as their services or actions are only triggered as a response to the actions of a primary actor, particularly the voters.

Use Cases



VOTER.

Log In. A voter can log into the system using their UP email. After inputting their credentials, the system will send a request to the authenticator (Google Authenticator) for 2-factor

authentication. Once the voter successfully accomplishes this, they will be logged into the system.

Cast Ballot. During the election proper, a voter can fill out their ballot and cast it. Some subroutines of this case include a display of a summary of their ballot to allow them to review it prior to casting, and the encryption of the ballot, handled by the backend (CodeIgniter). Finally, the submission of the ballot will also send a request for a reCAPTCHA. Only once the voter passes the reCAPTCHA will their vote be stored in the database.

View Ballot Receipt. Following their voting, a voter may log into the site in order to view their ballot receipt containing a summary of their vote.

View Election Results. A voter may track the results of the election in real time by viewing the election results on the site.

ADMIN.

Log In. An admin account may log into the system in order to edit or manage elections.

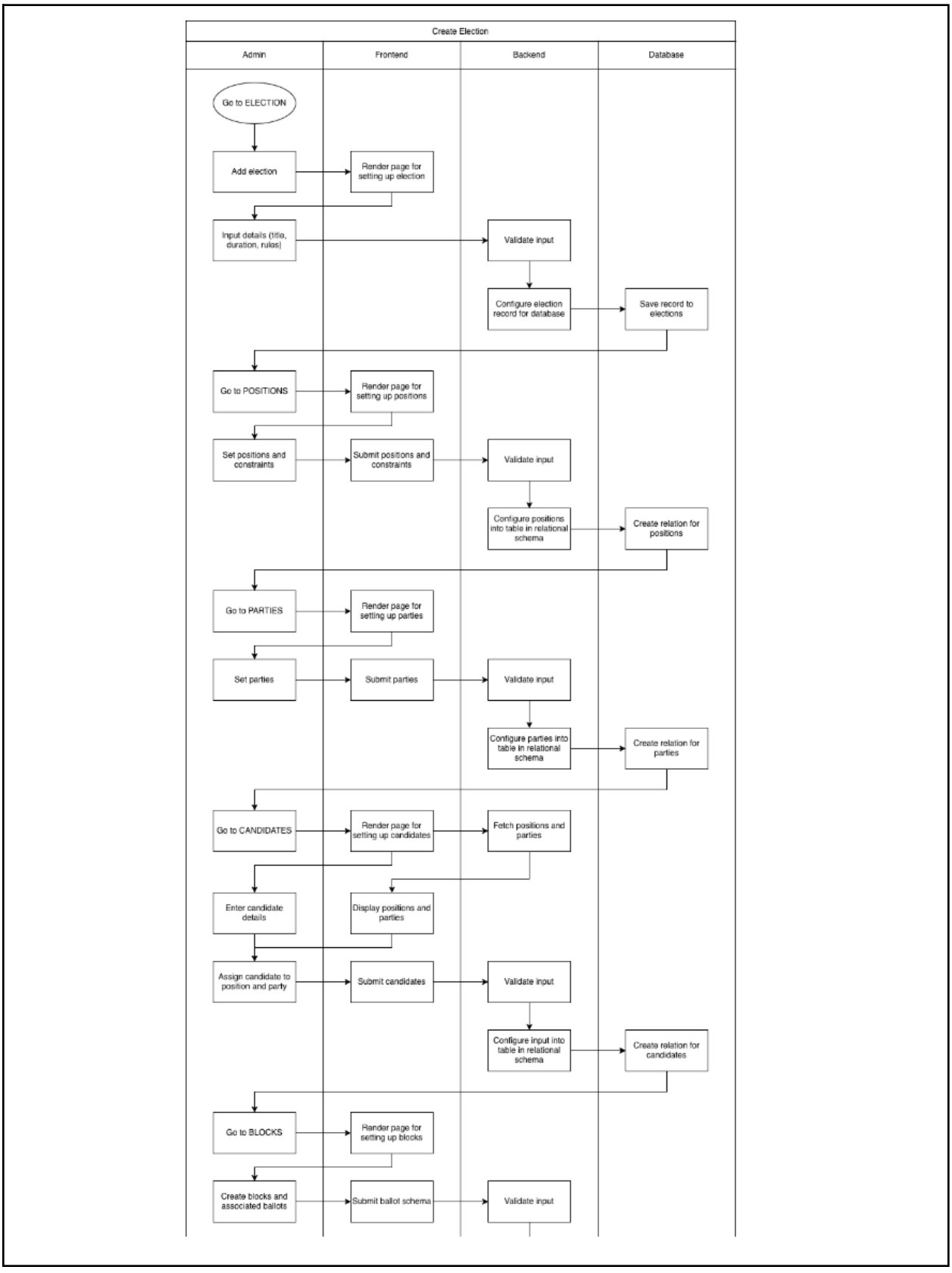
Create Election. An admin can create an election. Other than inputting the information and parameters for the election, other significant subroutines for the creation of elections are the creation of blocks which dictate the ballot of a certain subset of voters, the importing of the roster of registered voters, and the mapping of these voters to their respective blocks.

Start Election Proper. Once the election is scheduled to start, the admin must publish the election and set its status to “Active”. Afterwards, the system must disable admin permissions to edit any information or data related to the elections that might tamper with the results. This is a subroutine and the election can not proceed without this.

End Election Proper. Once the duration of the election has lapsed, an admin may officially end the election and set its status to “Inactive”. This will finalize all the results.

View Election Results and Statistics. Throughout the election and after it, an admin may view the running results and statistics.

System Sequence Diagrams



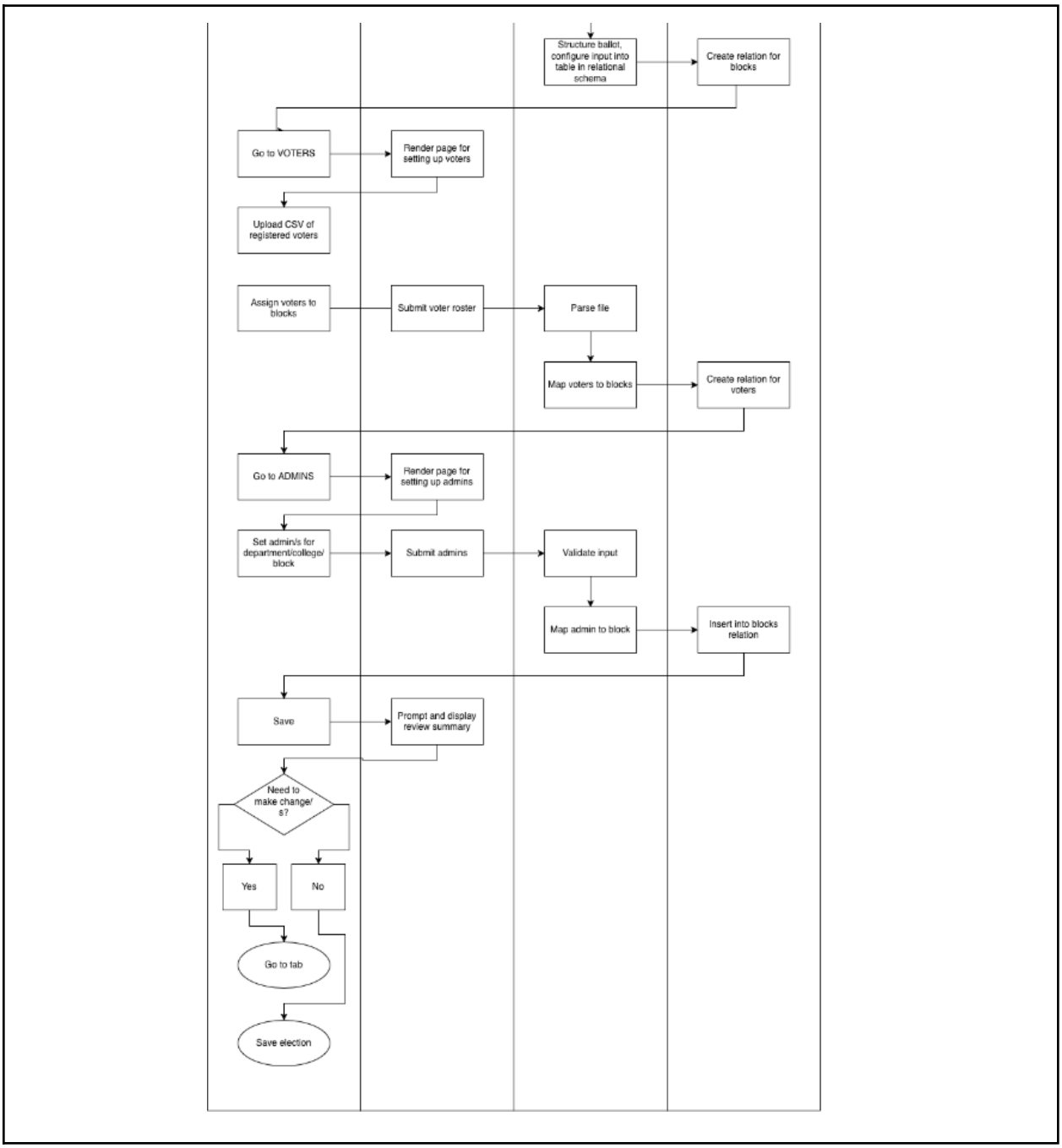


Figure X. Swimlane Diagram for Admin Use Case “Create Election”

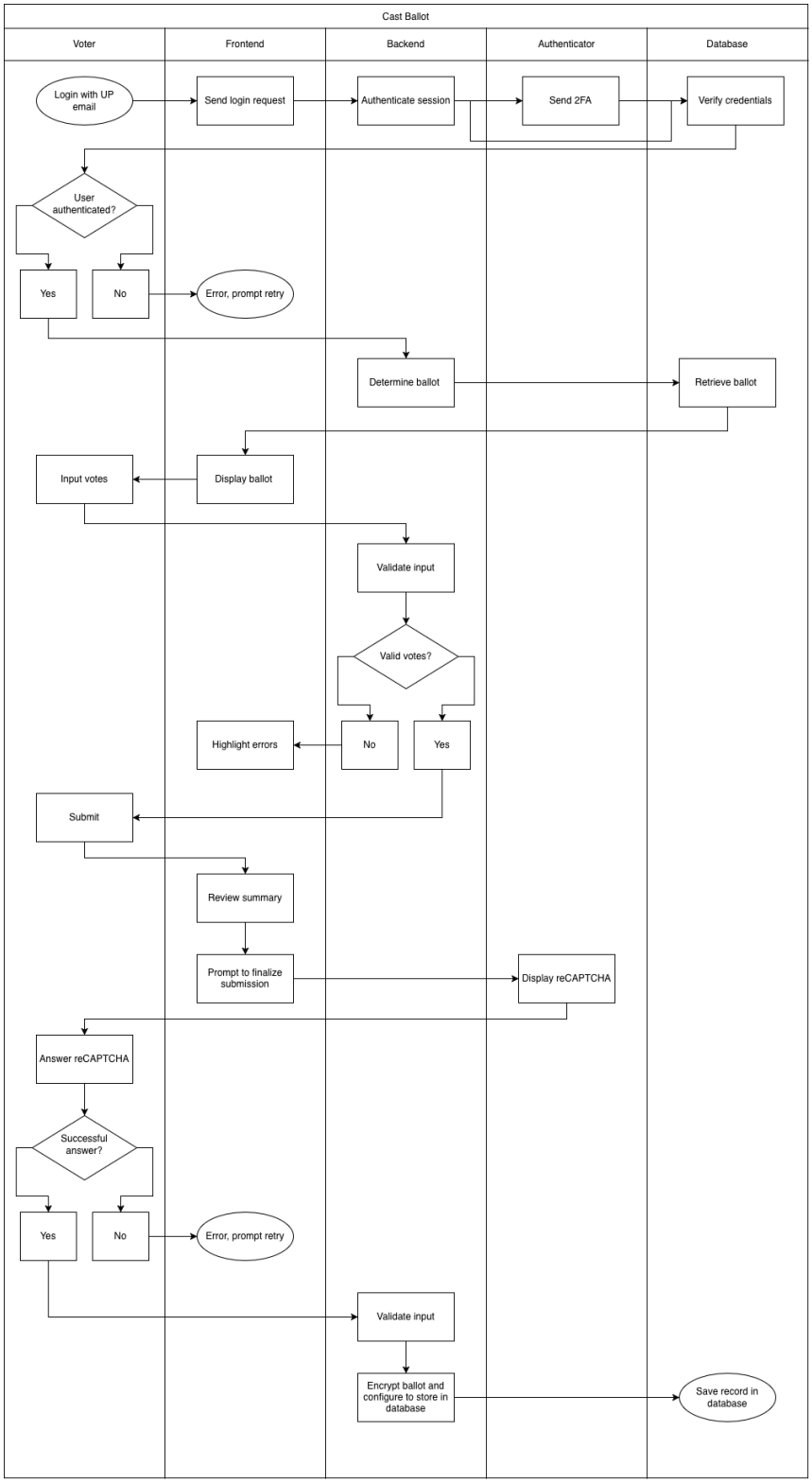


Figure X. Swimlane Diagram for Voter Use Case “Cast Ballot”

1.3 Configuration Model

This section specifies the allocation of software functionality and subsystems within the physical computing environment.

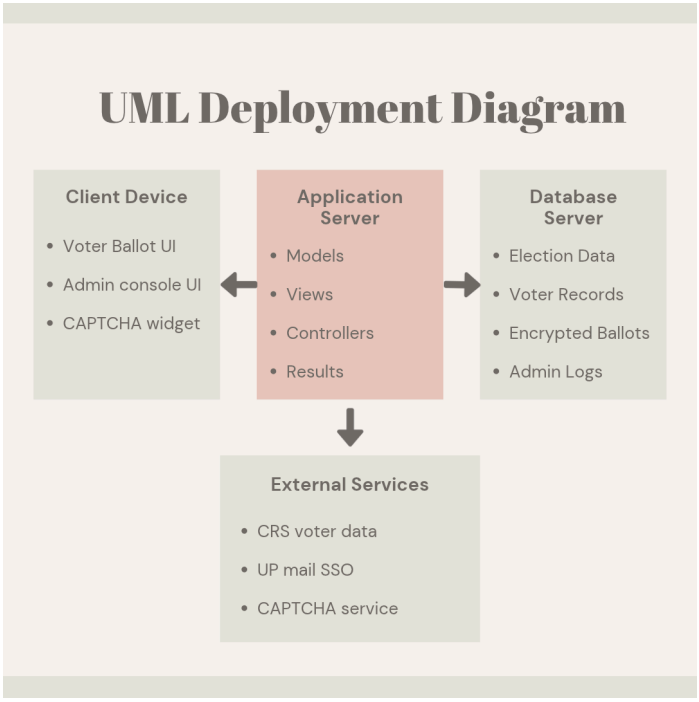


Figure 2. UML Deployment Diagram

The UML Deployment Diagram shown above has three main nodes: the client device, the application server, and the database server, with three external services needed below. The application server and database server can start on one VPS during development. They are separate nodes in the deployment model so the database can be isolated and scaled independently, and it should only accept connections from the application server.

References

Pressman, R. S., & Maxim, B. R. (2010). *Software engineering: A practitioner's approach* (7th ed.). R.S. Pressman & Associates, Inc.